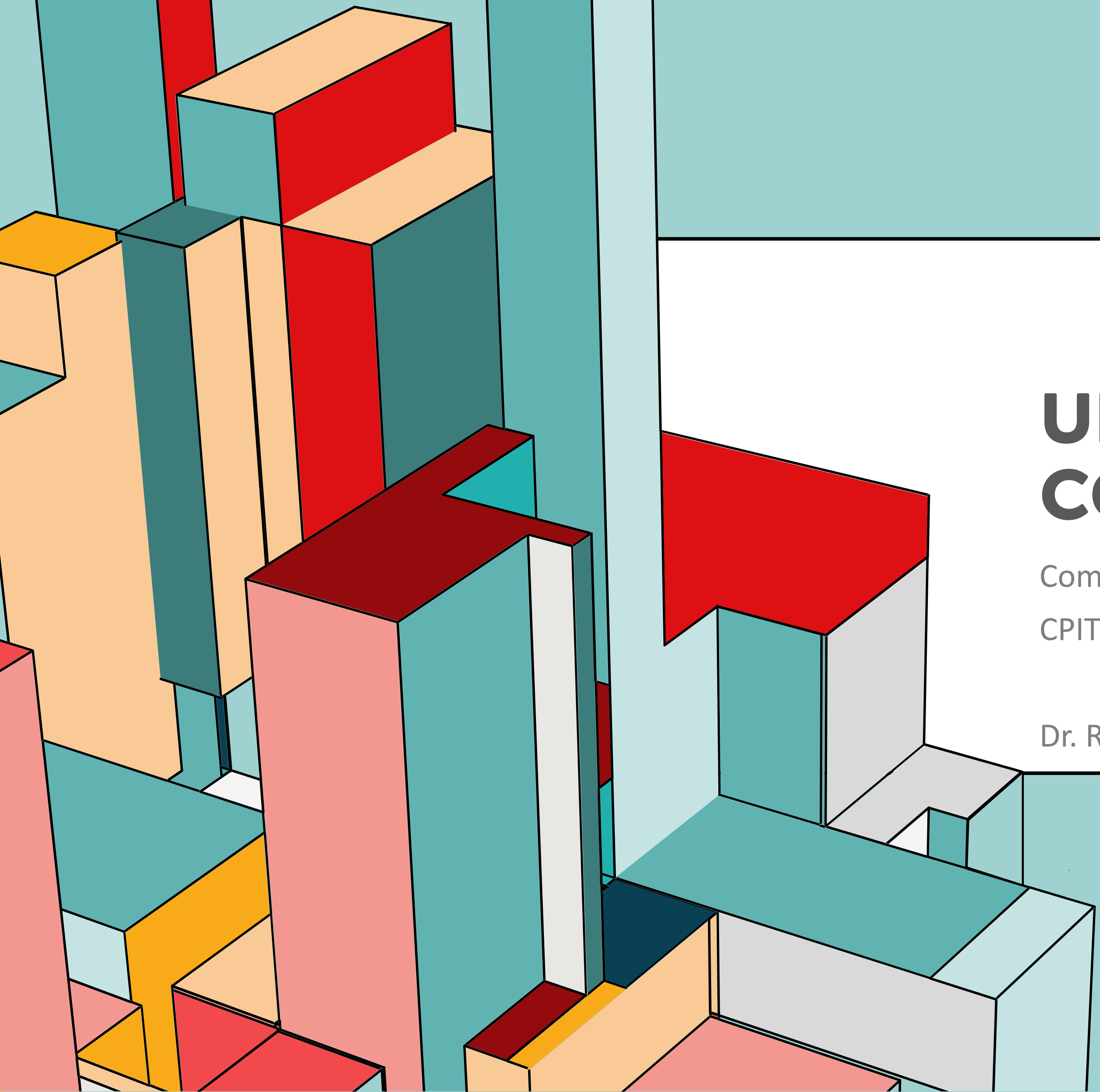


UNDERSTANDING COLORS

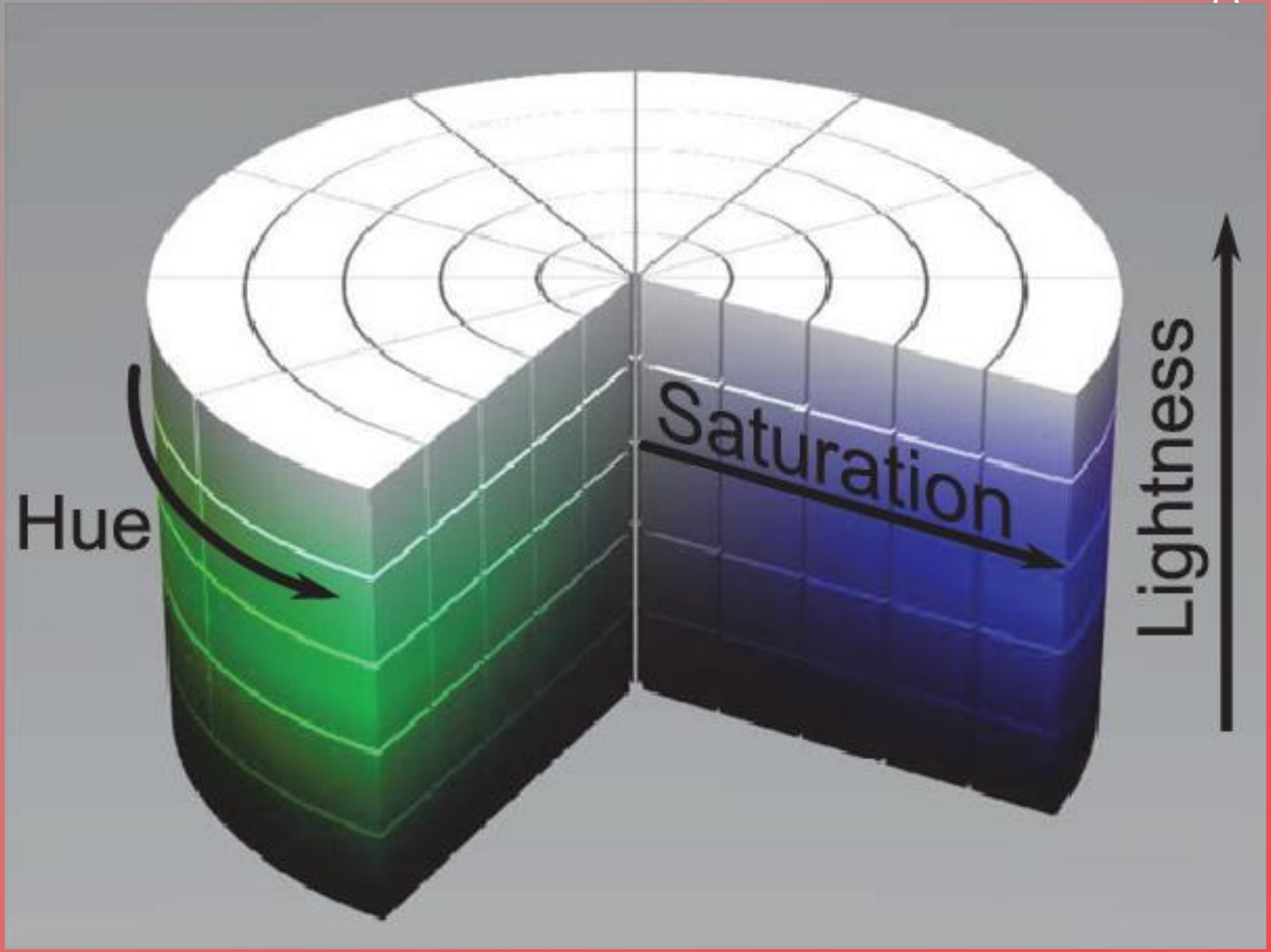
Computer Graphics

CPIT285

Dr. Rayed AlGhamdi

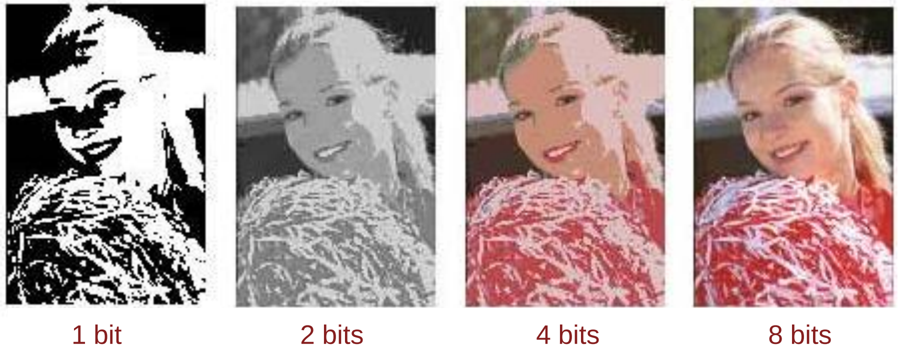
COLORS MODELS

How does the computer work with colors?



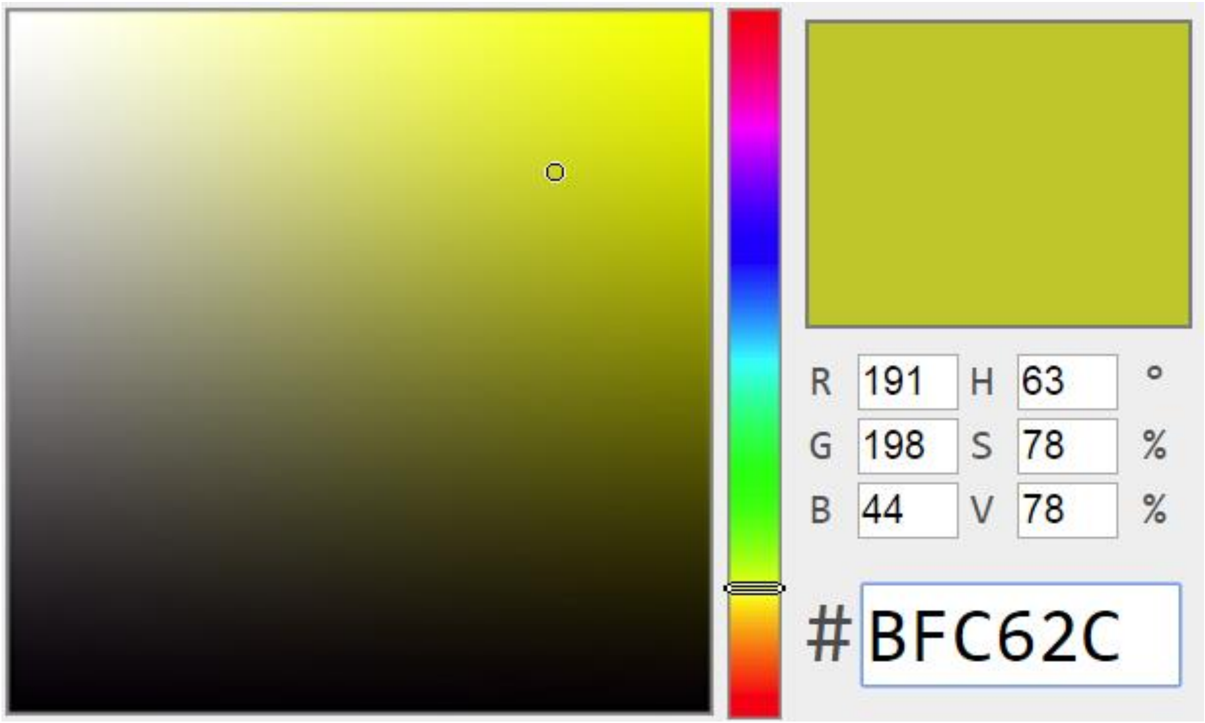
RGB COLORING MODEL

- **Color depth** depends on the number of bits used to represent each pixel, which is determined by the encoding systems the computer uses.
- A common use for coloring is 8 bits representing the value of each of the three main colors RGB: Red, Green & Blue (8*3=24 bits)
- If each of the three main colors uses 8 bits (24 bits total), how many colors can be generated using 24 bits?
  - $2^{24} = 16{,}777{,}216$



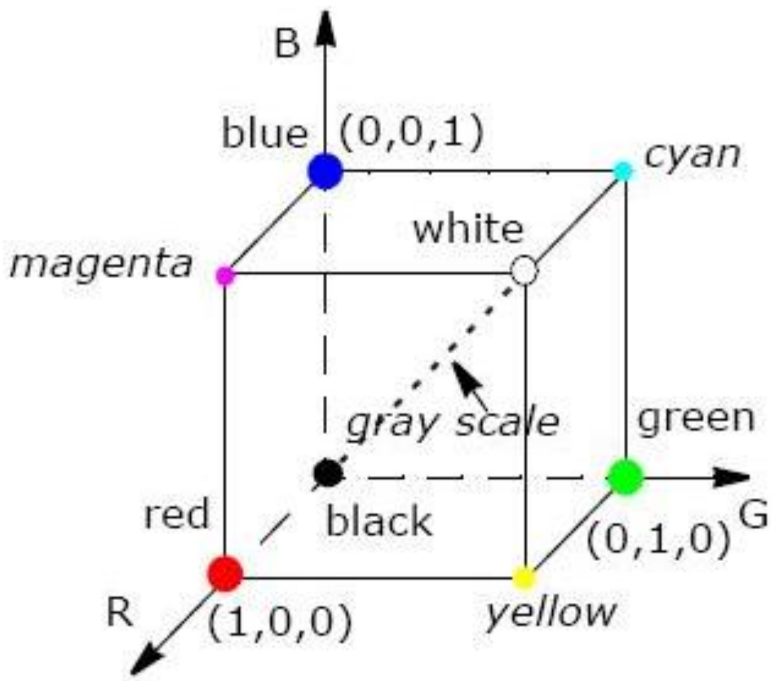
RGB COLORING MODEL

- Try this link to check the value of each color
- What do these values mean to you?
- Are they codes used for coloring?
- How to find the appropriate code for your application?



RGB COLORING MODEL

- In OpenGL, each color in RGB model is identified using a value 0.00 to 1.00
- Each main color can take 100 different values (0.0, 0.01, 0.02, …. 0.99, 1.0)
  - ◆ = glColor3f(1.0, 0.0, 0.0)
  - ◆ = glColor3f(0.0, 1.0, 0.0)
  - ◆ = glColor3f(0.0, 0.0, 1.0)



RGB COLORING MODEL

- Values in the range 0 to 255 can be used too…
  - ◆ = glColor3ub(255, 0, 0)
  - ◆ = glColor3ub(0, 255, 0)
  - ◆ = glColor3ub(0, 0, 255)
  - ◆ = glColor3ub(0, 0, 0)
  - ◆ = glColor3ub(255, 255, 0)

    ub → unsigned byte
- RGB model is an additive coloring: adding colors to each other to generate more colors…

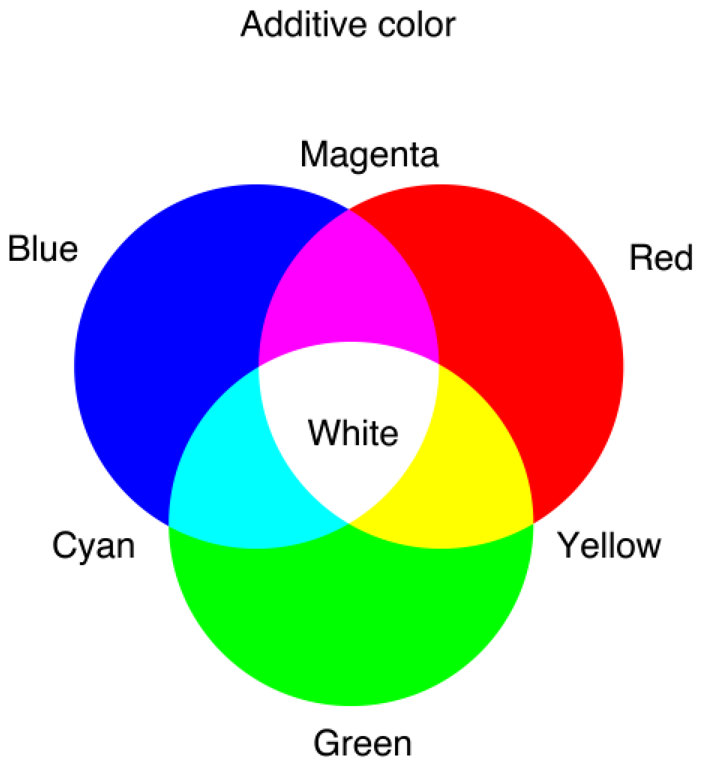
CMYK COLORING MODEL

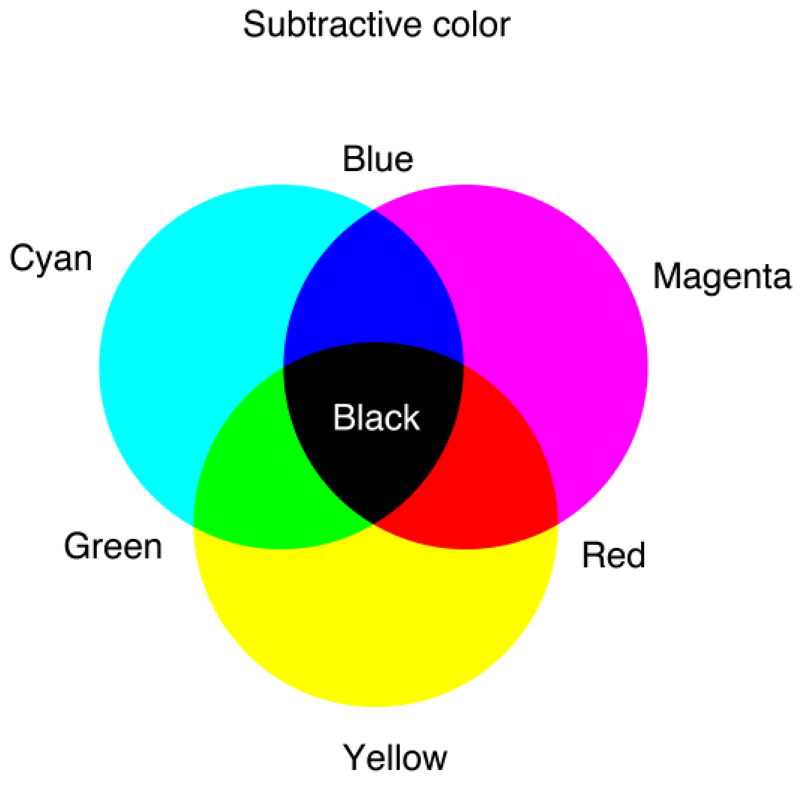
- Values in the range 0 to 255 can be used too…
- It uses Cyan, Magenta, Yellow & Black
- Mainly used for printers
- It is a subtractive coloring
- It starts with light/ white and subtracts colors to generate new colors…

RGB vs CMYK

What is the difference between additive and subtractive coloring systems?

RGB





CMYK

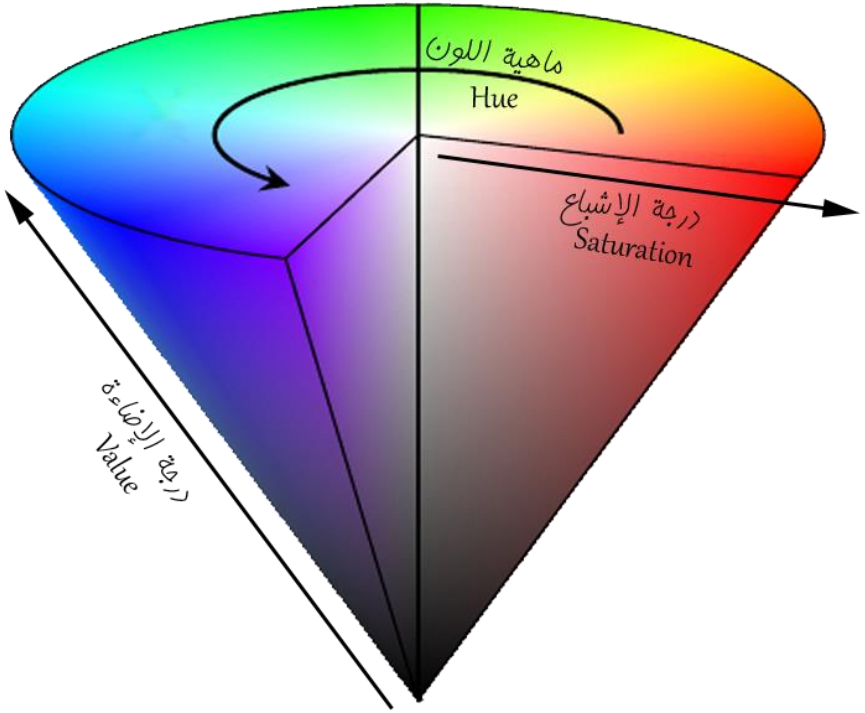
RGB vs CMYK

What is the difference between additive and subtractive coloring systems?

- Additive color is one created by mixing red, green, and blue light in different combinations.
- Additive colors begin as black and become brighter as you add different light.
- In contrast, a subtractive color is made by partial absorption of different colors of paint or ink.
- Subtractive colors begin as white and take on the appearance of the added colors or their mixtures.

HSV/HSL COLORING MODEL

- Developed to represent very close to real colors based on the RGB coloring scheme.
- Each color is identified based on 3 items:
  - **H**ue (0-360)
  - **S**aturation (0-100)
  - **V**alue/**L**ightness (0-100)

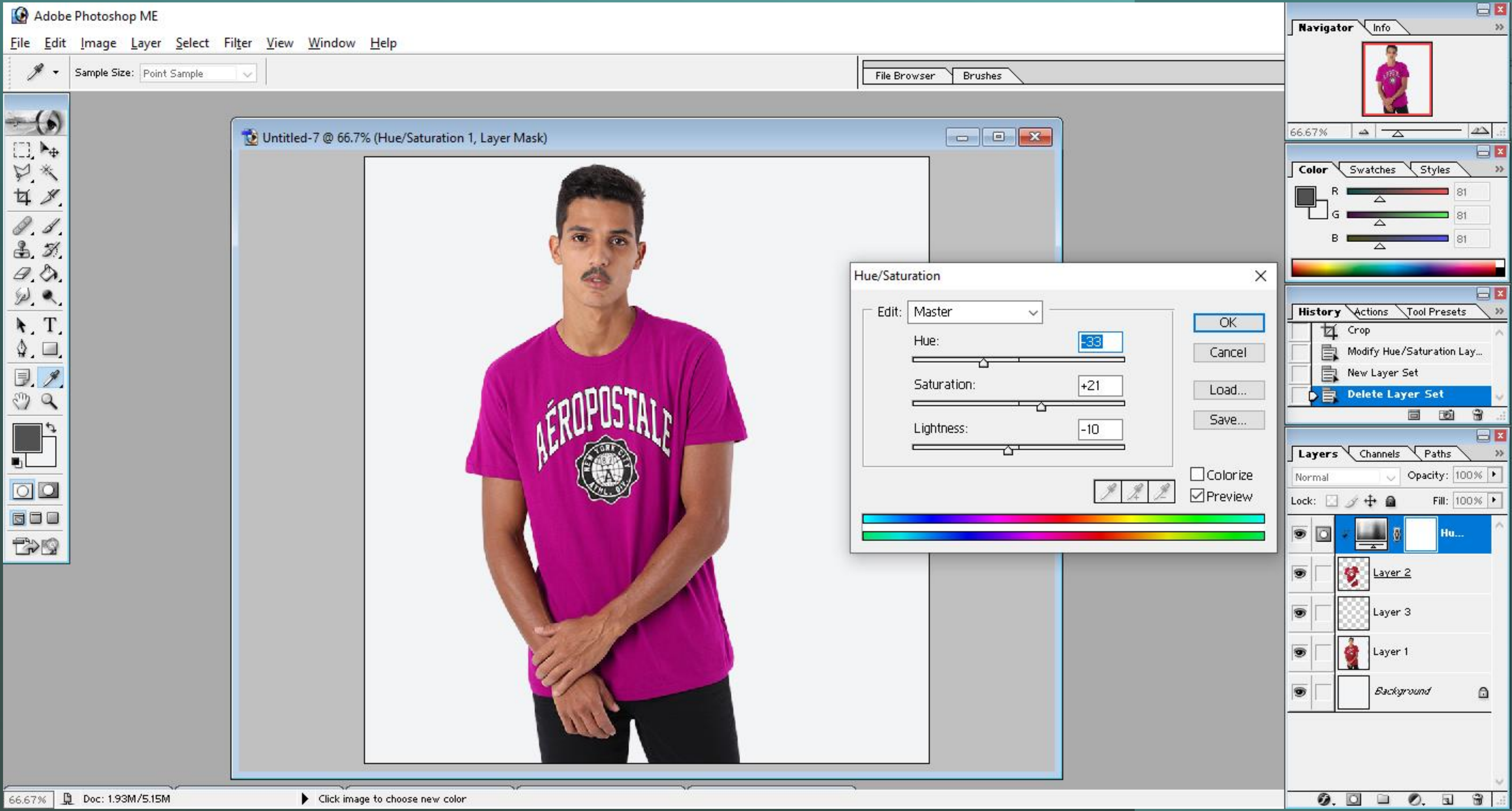


HSV/HSL COLORING MODEL

- Hue of a color refers to which pure color it resembles. All tints, tones, and shades of red have the same hue. It takes a value of 0 - 360
- Saturation of color describes how white the color is. A pure red is fully saturated, with a saturation of 1; tints of red have saturations less than 1, and white has a saturation of 0.
- Value of color, also called its lightness, describes how dark the color is. A value of 0 is black, with increasing lightness moving away from black.
- Check out [HSV Color Picker]

HSV/HSL COLORING MODEL

- The use of this coloring model gives the user more flexibility to play with colors.
- Go ahead with your personal photo wearing a colored T-shirt and change the color by playing with the three elements of HSV: Hue, Saturation, and Value.
- This task can be easily achieved using photoshop. If you do not have photoshop, you may use Krita (open-source software).


Adobe Photoshop ME
File Edit Image Layer Select Filter View Window Help
Sample Size: Point Sample
File Browser
Brushes
Untitled-7 @ 66.7% (Hue/Saturation 1, Layer Mask)
AÉROPOSTALE
Hue/Saturation
Edit: Master
Hue: -33
Saturation: +21
Lightness: -10
OK
Cancel
Load...
Save...
Colorize
Preview
Navigator
Info
66.67%
Color
Swatches
Styles
R 81
G 81
B 81
History
Actions
Tool Presets
Crop
Modify Hue/Saturation Lay...
New Layer Set
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Layers
Channels
Paths
Normal
Opacity: 100%
Lock:
Fill: 100%
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Layer 2
Layer 3
Layer 1
Background
66.67%
Doc: 1.93M/5.15M
Click image to choose new color

COLOR WHEEL

How does the computer work with colors?

TRADITIONAL COLORING THEORY

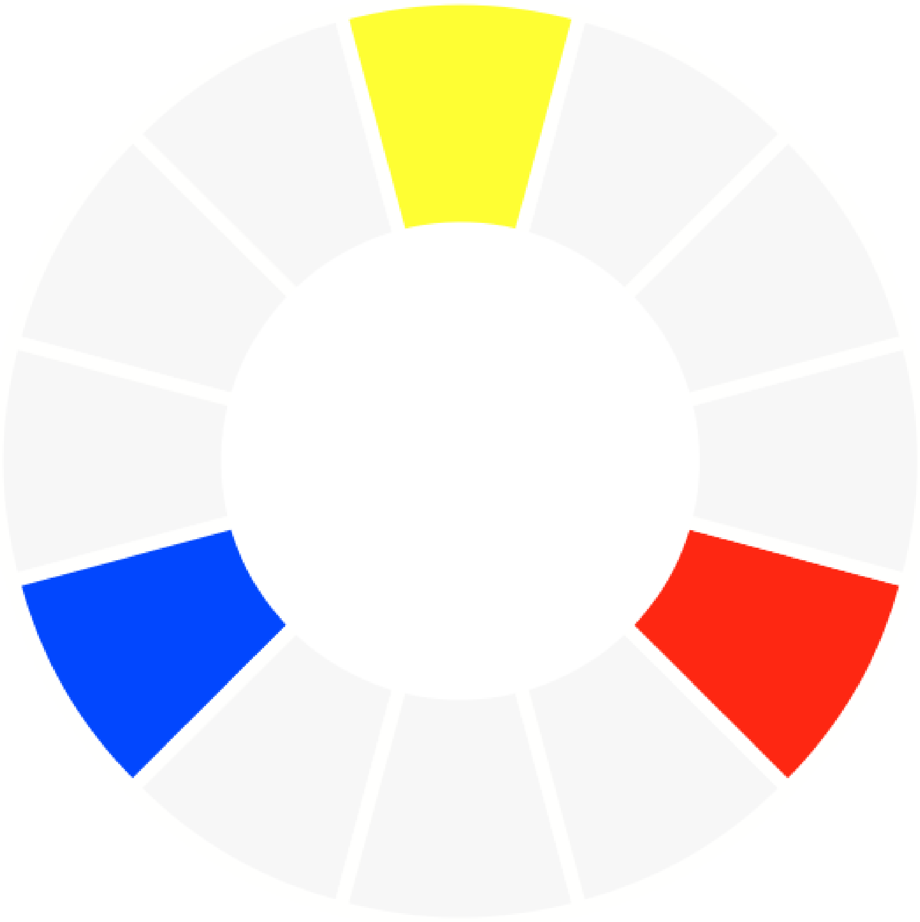
COLORS

- Very early in school, we may start using 12 colors
- We learn the 12 colors are made of three primary colors

TRADITIONAL COLORING THEORY

COLORS

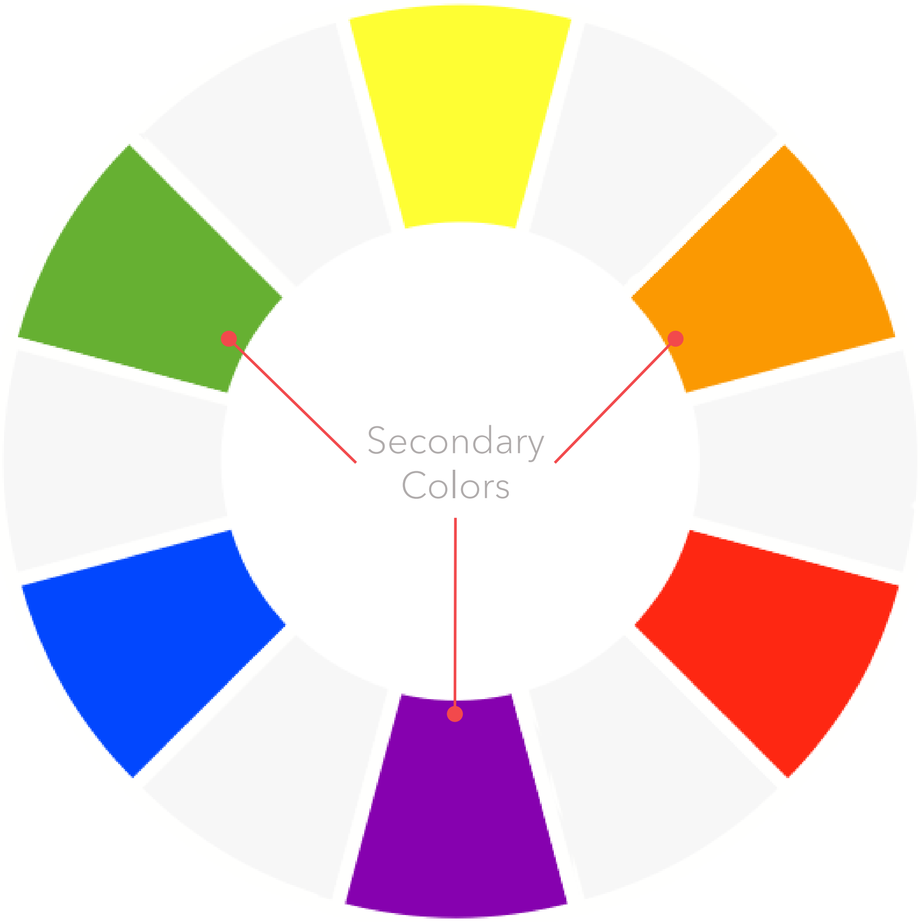
- Primary Colors
  - Red
  - Blue
  - Yellow



TRADITIONAL COLORING THEORY

COLORS

- By mixing primary colors three more colors can be generated, called secondary
  - Orange
  - Green
  - Violet



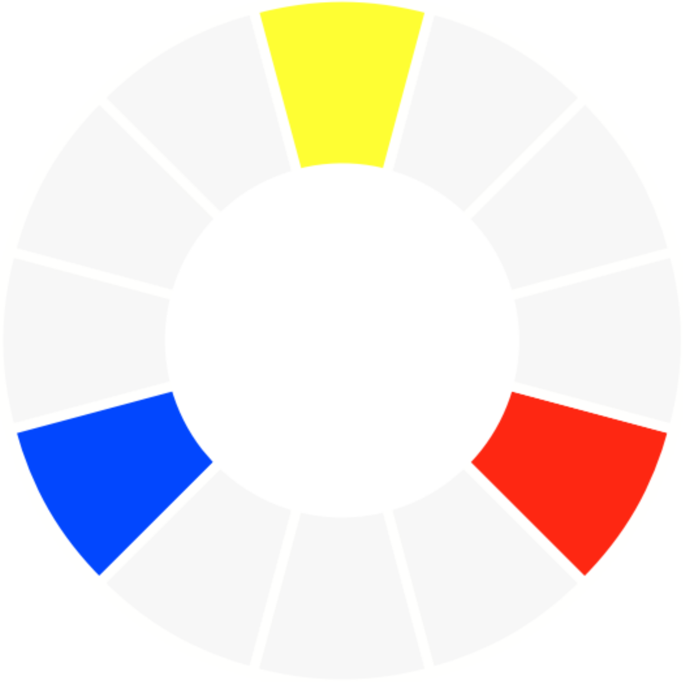
TRADITIONAL COLORING THEORY

COLORS

- Furthermore, mixing a primary color with a secondary, generate six more colors called tertiary:
  - Yellow-orange
  - Red-orange
  - Red-violet
  - Blue-violet
  - Blue-green
  - Yellow-green

TRADITIONAL COLORING THEORY

COLORS



Primary Colors

Secondary Colors

Tertiary Colors

TRADITIONAL COLORING THEORY

COLOR WHEEL

- The combination of the primary, secondary, and tertiary colors consists of 12 colors.
- These 12 colors are what’s called “color wheel”.
- The **color wheel** is an illustrative organization of color hues around a circle, which shows the relationships between primary colors, secondary colors, tertiary colors.

TRADITIONAL COLORING THEORY

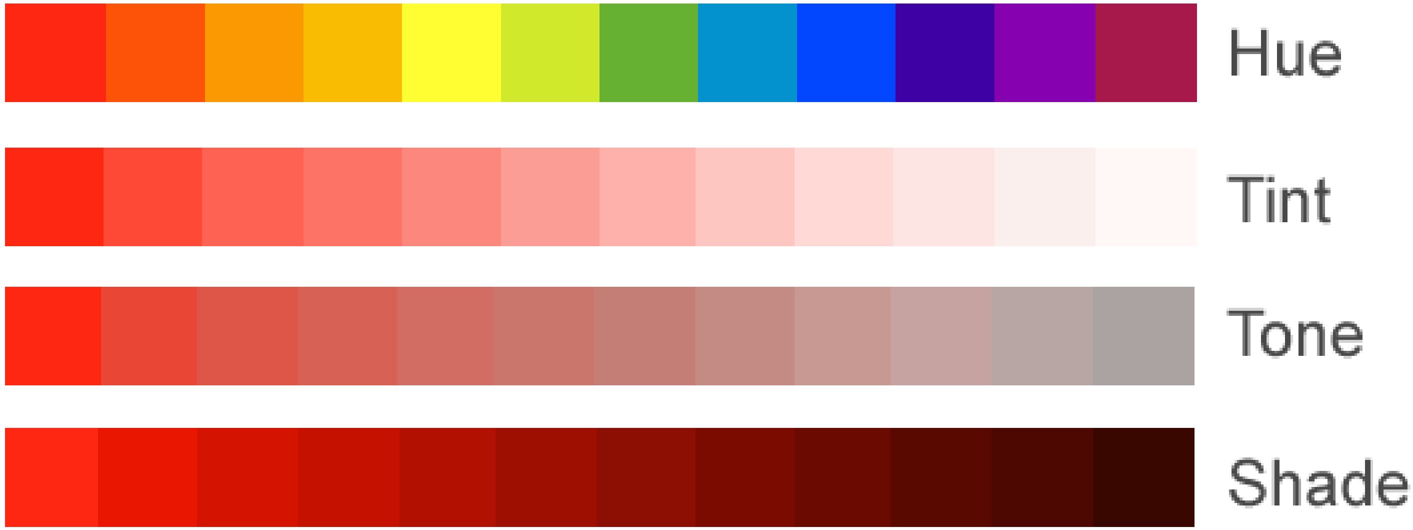
COLOR WHEEL

- More colors can be generated by further mixing these 12 color hues that make up the color wheel.
- How many colors can you recognize in this circle?
- Let's say: 100 different colors.
- Can more colors be generated?

TRADITIONAL COLORING THEORY

COLOR WHEEL

- More colors can be generated by playing with each color hue's:
  - Tint;
  - Tone; and
  - shade



TRADITIONAL COLORING THEORY

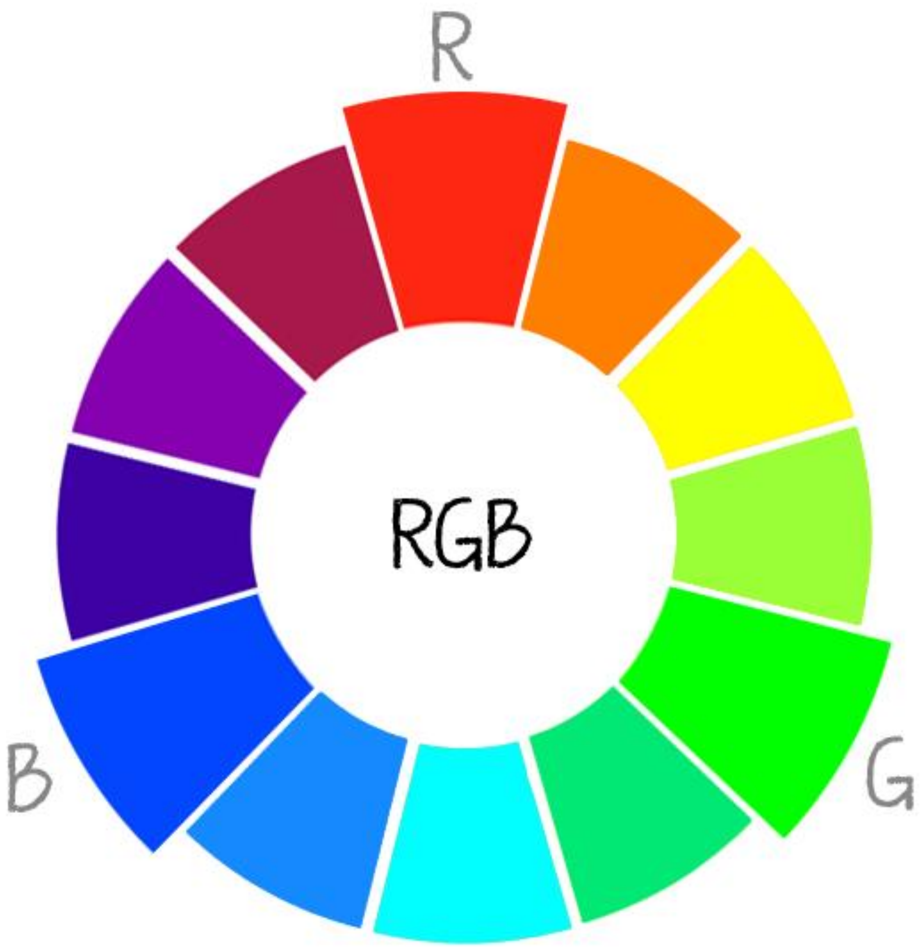
COLOR WHEEL

- **Hue** refers to the origin of the color we use →

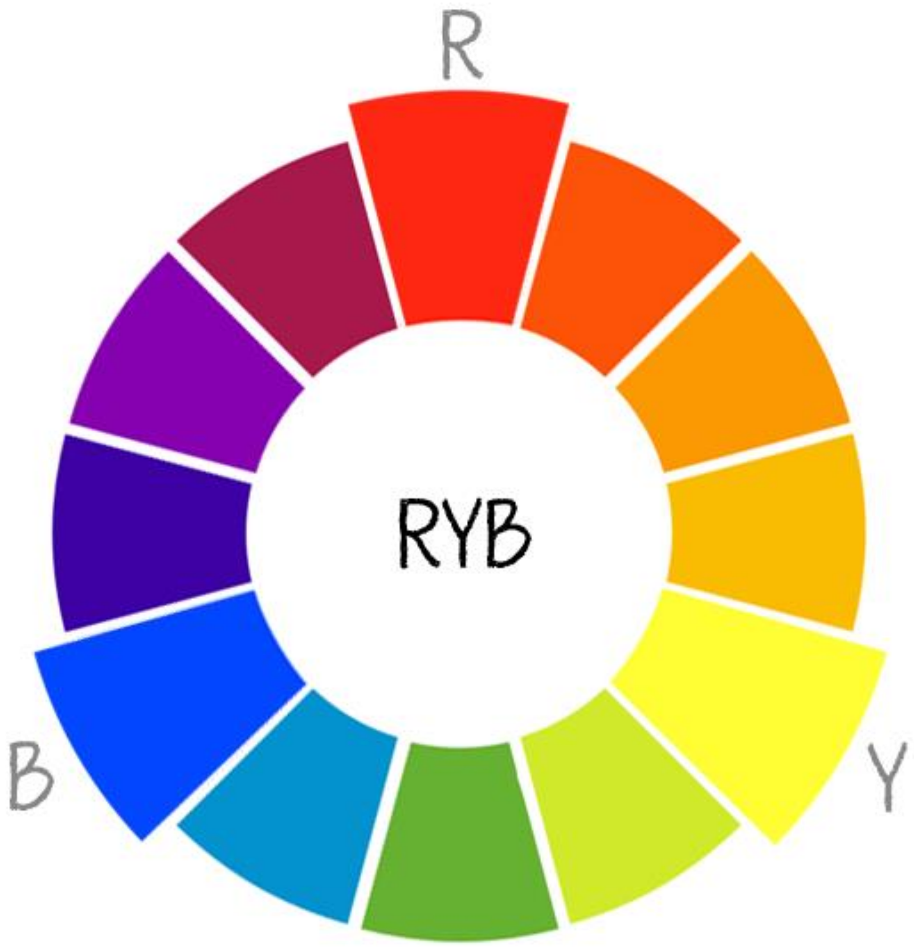
- **Tint:** a mixture of pure colors with only White added

- **Tone:** a mixture of pure colors with only Gray added

- **Shade:** a mixture of pure colors with only Black added



Additive Coloring System

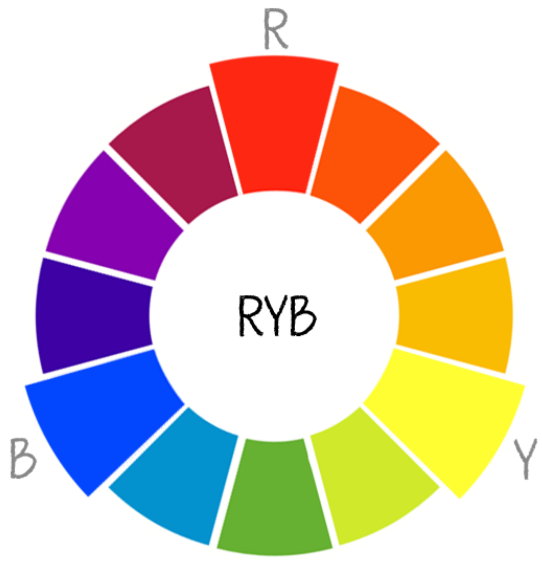


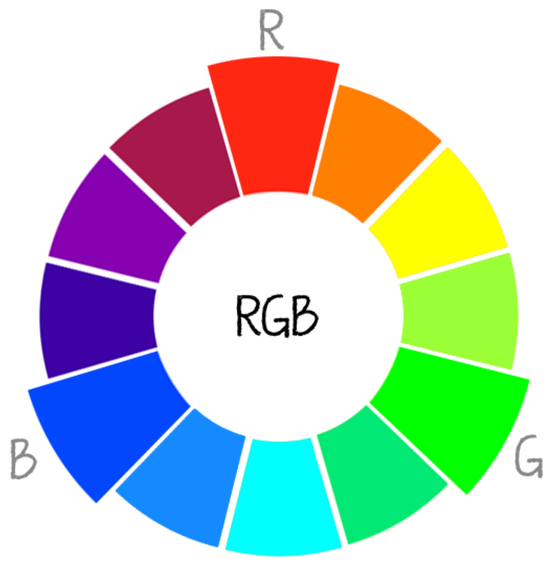
Subtractive Coloring System

PAY ATTENTION

What are the real primary colors?

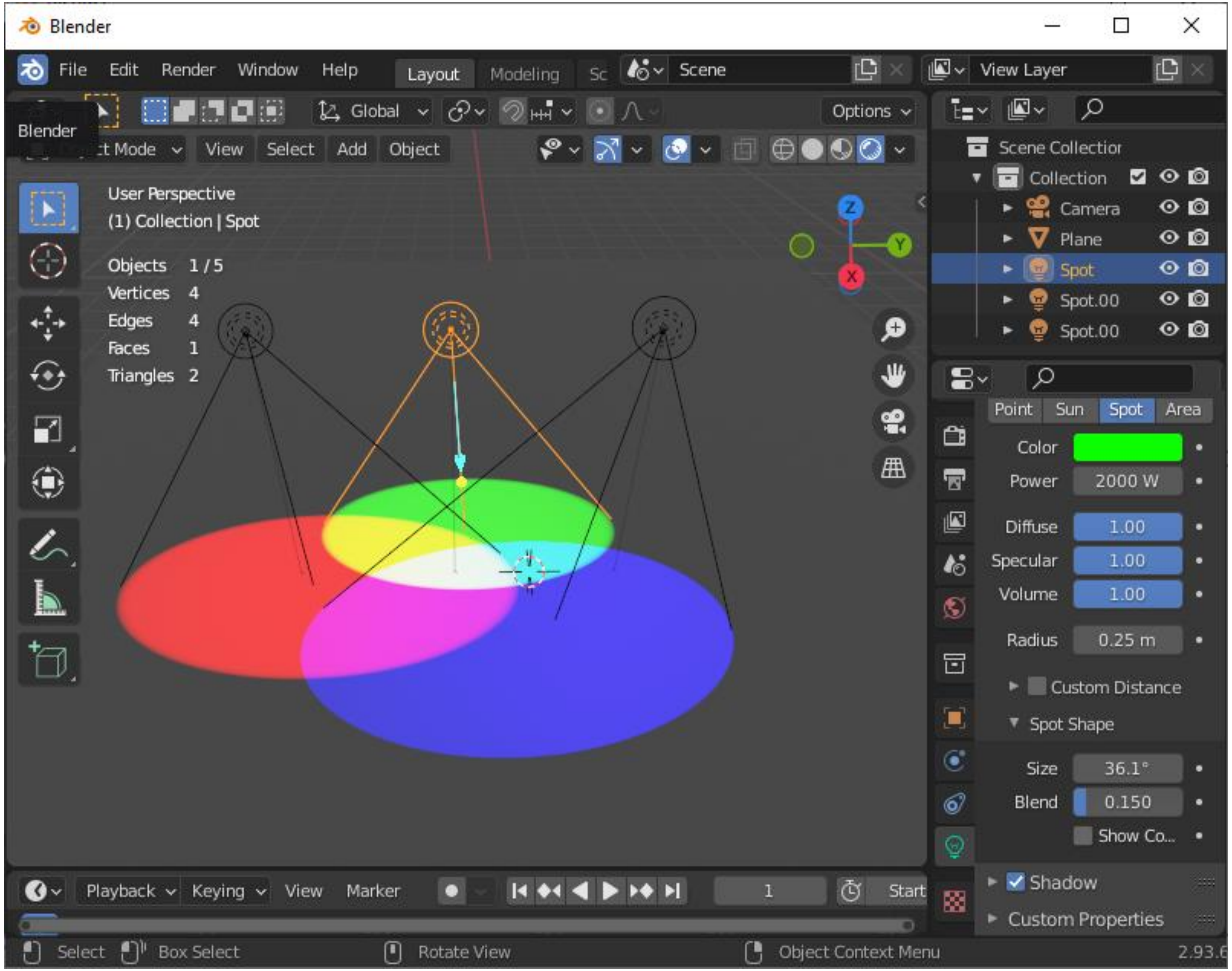
- In art class, we learned that the three primary colors are red, yellow, and blue. Subtractive coloring system.
- In the world of physics and light, however, the three primary colors are red, green, and blue. Additive coloring system.
- Additive colors are those which make more light when they are mixed together.
- Display screens started with black, light is used, and that's why an additive coloring system is used.





TRY IT YOURSELF

- Add three spotlight sources on blender
- Color them red, green, and blue
- Make the light sources intersect each other on a white plane
- What is the color in the intersection point of the three colors?



DISPLAY SCREEN

- Keep in mind, display screen work with light
- Colored lights are used, and it starts with black
- It is different from paint and ink that start with white

COLORING HARMONY

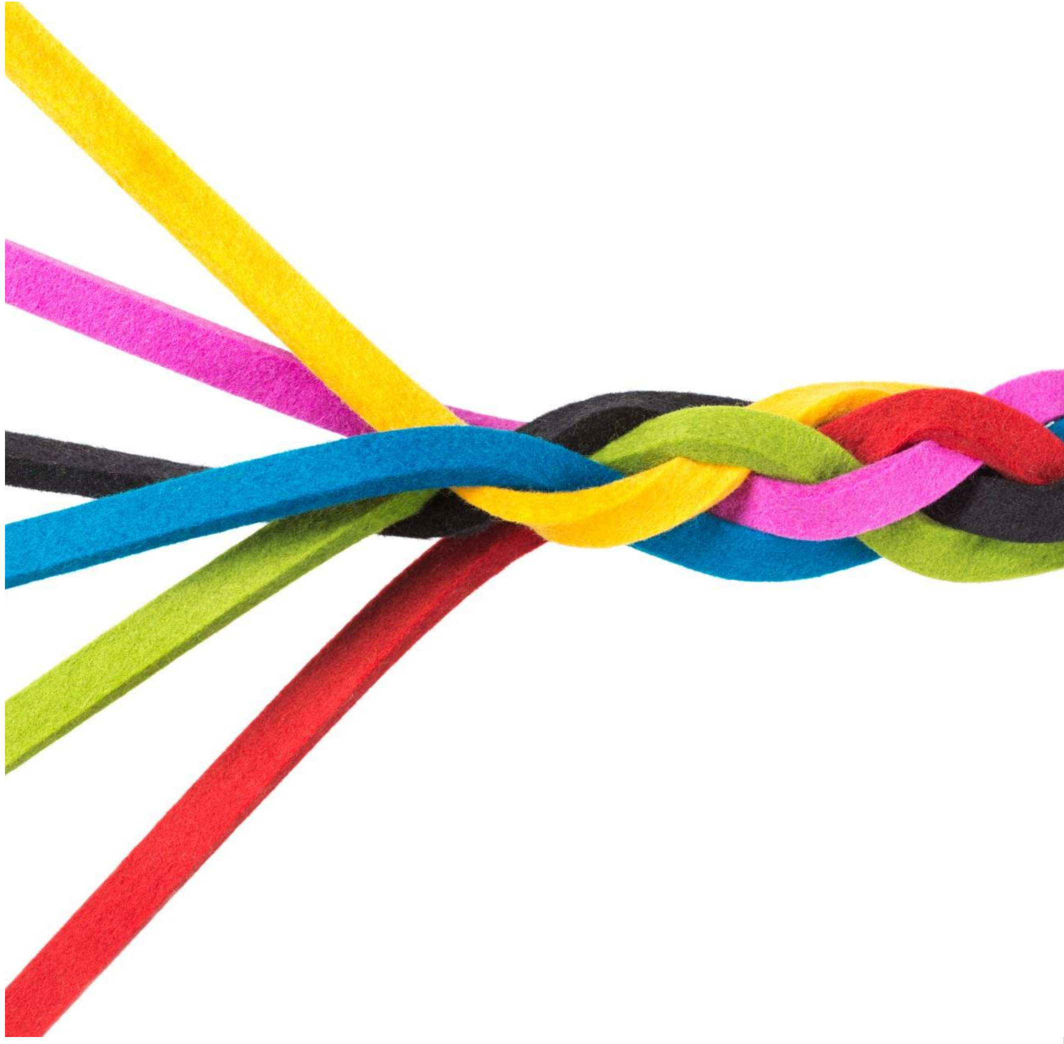
It refers to aesthetically pleasing and harmonious color combinations based on geometric relationships on the color wheel.

COLOR HARMONY

- Thus far, how these colors can be used together?
- Some combinations of colors are pleasing to the eye and some others are not!
  - Are there ways to choose/organize colors to be eye-pleasing?
  - How do these ways help?

COLOR HARMONY

- Harmony can be defined as a pleasing arrangement of parts.
- In visual experiences, harmony is something that is pleasing to the eye.
- When something is not harmonious, it's either boring or chaotic.
- So how do we achieve color harmony, using hue, saturation and value ranges?

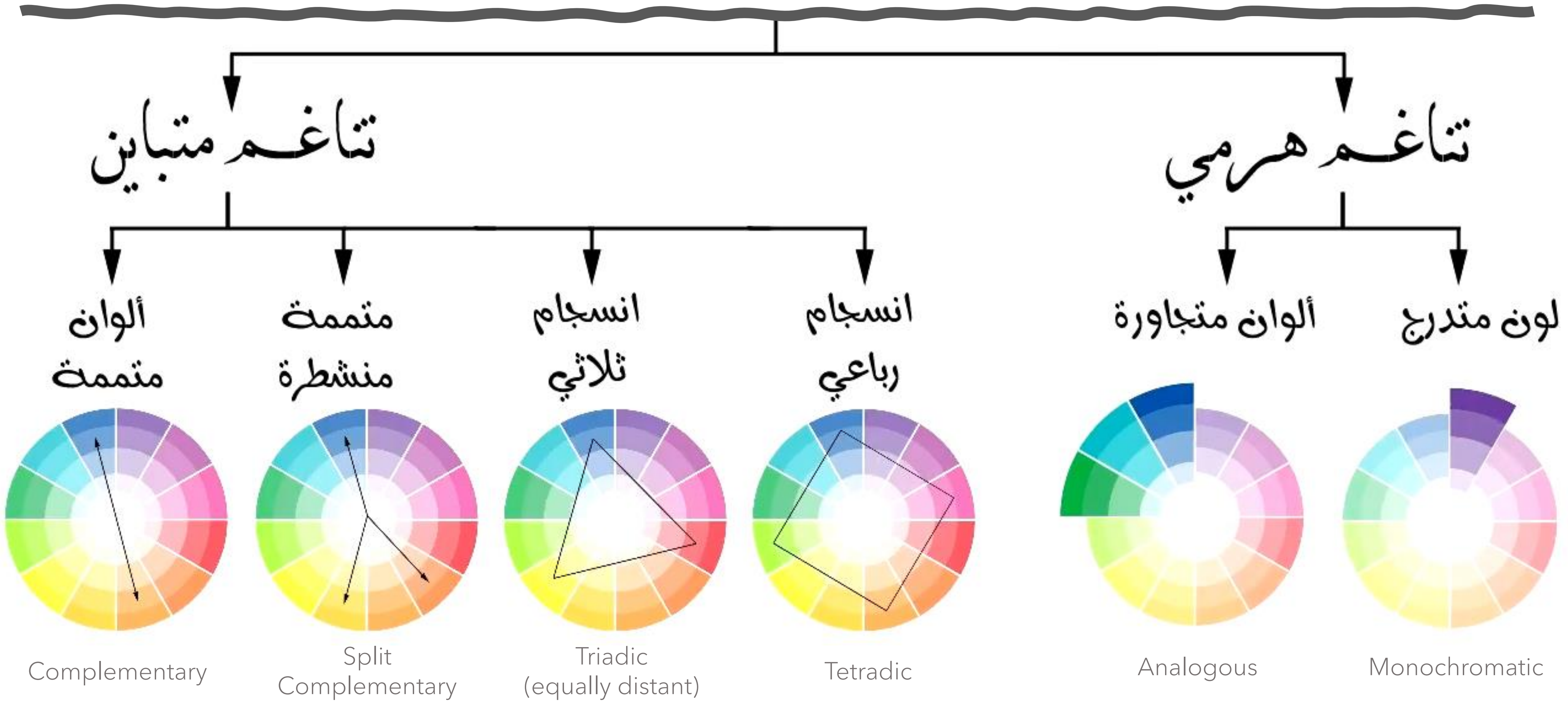


COLOR HARMONY SCHEMES

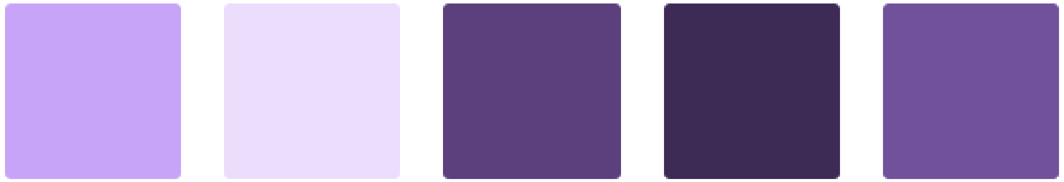
Coloring harmony schemes can be divided into two main groups (each consist a number of schemes):

- Gradual Harmony
  - Monochromatic
  - Analogous
- Contrastive Harmony
  - Complementary
  - Split complementary
  - Triadic (equally distant on the wheel)
  - Tetradic (double complementary)

COLOR HARMONY SCHEMES

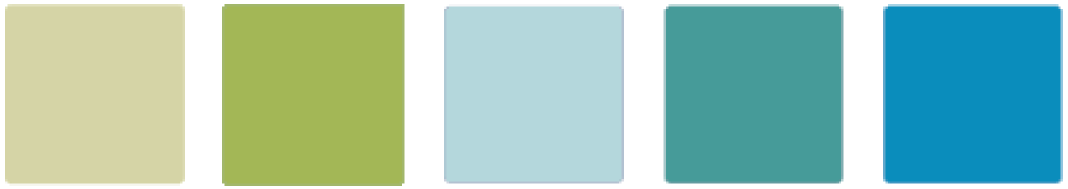


MONOCHROMATIC COLORING SCHEME



- This is the easiest style for harmonic palette
- It uses only one color from the color wheel
- Variations can be created by using *saturation* and *value*

ANALOGOUS COLORING SCHEME



- It uses the colors which are next to each other in the color wheel, such as from blue to green.
- Playing with these you can get a variety of well-mixed colors for your design.

COMPLEMENTARY COLORING SCHEME

- It uses colors opposite from one another in the color wheel, such as blue and orange.
- To create an interesting and refreshing palette, we again use lighter or darker shades of colors & a different intensity.

SPLIT COMPLEMENTARY COLORING SCHEME

- It uses the two colors adjacent to its complement.
- This color scheme has the same strong visual contrast as the complementary color scheme, except we have more colors to play with.

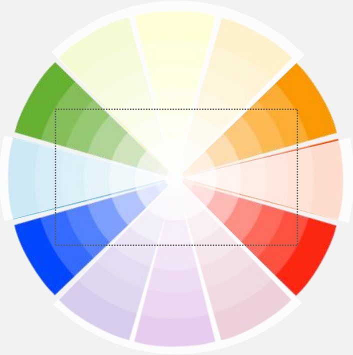
TRIADIC COLORING SCHEME

- It is to choose ones that are equally spaced on the color wheel.
- The combinations are visually very effective.
- Use with caution!
- One color to be dominated and use the two others for accent.

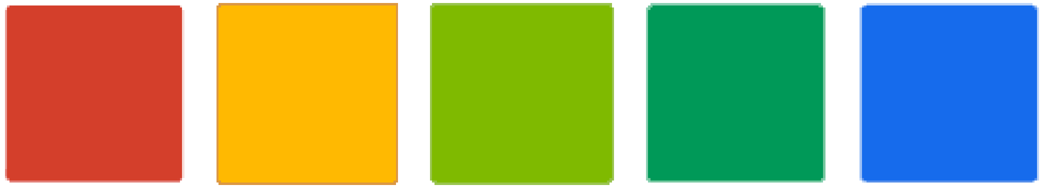
TETRADIC COLORING SCHEME

- It can be called double complementary.
- Color mixing is using four colors from the wheel.
- The selected colors make a rectangular or square shape on the wheel.
- The palette looks best if one color dominates, and others are used for details and as refreshment.

TETRADIC COLORING SCHEME



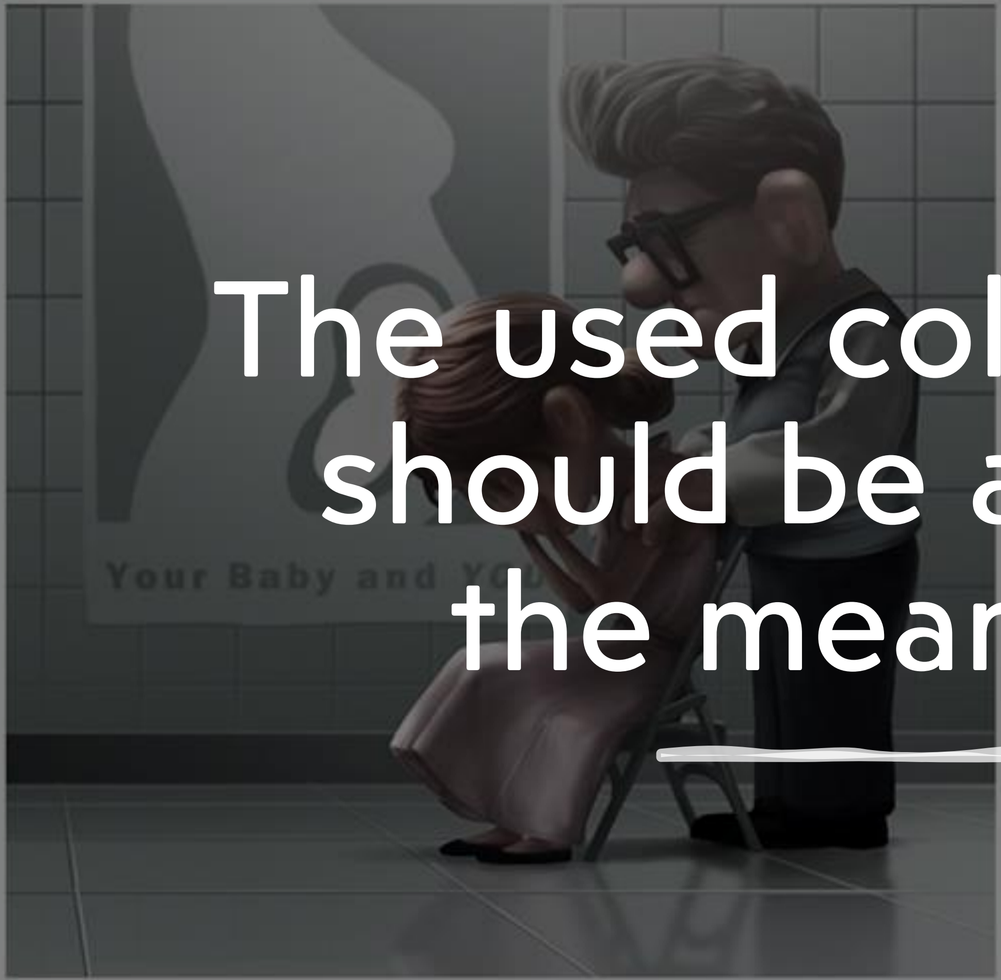
- Tetradic coloring scheme is commonly used in popular logos...
- Pay high attention to the equal usage of colors!
- Equal usage should be OK with small scale such as logos, but you need to be careful when applying on a large scaling.



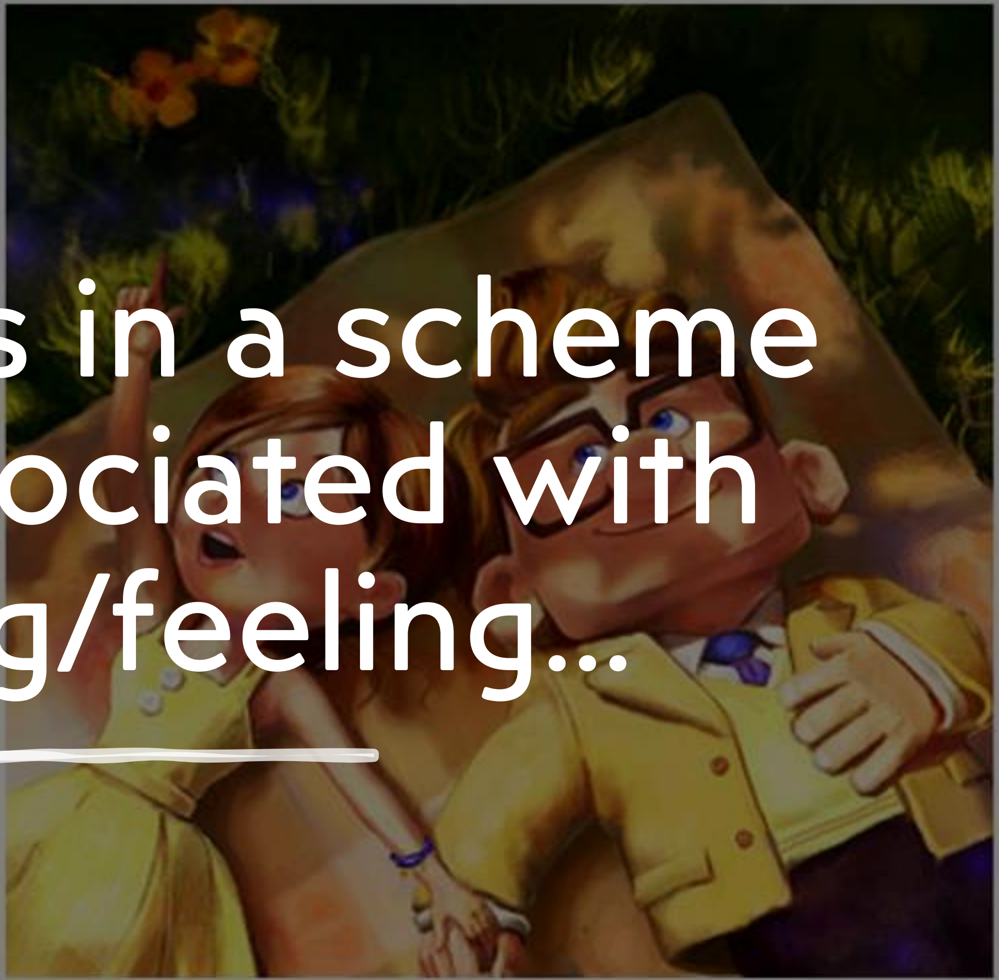
MEANINGS & FEELINGS OF COLORS

Effective coloring usage should communicate with the viewers to share the same meaning or feeling that a designer wanted to reach.

The used colors in a scheme should be associated with the meaning/feeling…



تناغم ألوان هرمية مع تقليل درجة الإشباع
للتوافق مع معنى الحزن



تناغم ألوان متباينة مع زيادة درجة الإشباع
للتوافق مع معنى السرور

Cool Colors

Cool colors tend to have a calming effect on the viewer. Used alone however, these colors can have a cold or impersonal feel, so when choosing cool colors, it may be wise to add a color from another group to avoid this and add warmth to your palette.

Warm Colors

Warm colors tend to have an exciting effect. However, when these colors are used alone they can over-stimulate, generating emotions of anger and violence. When choosing warm tones, adding colors from another group will help to balance this.

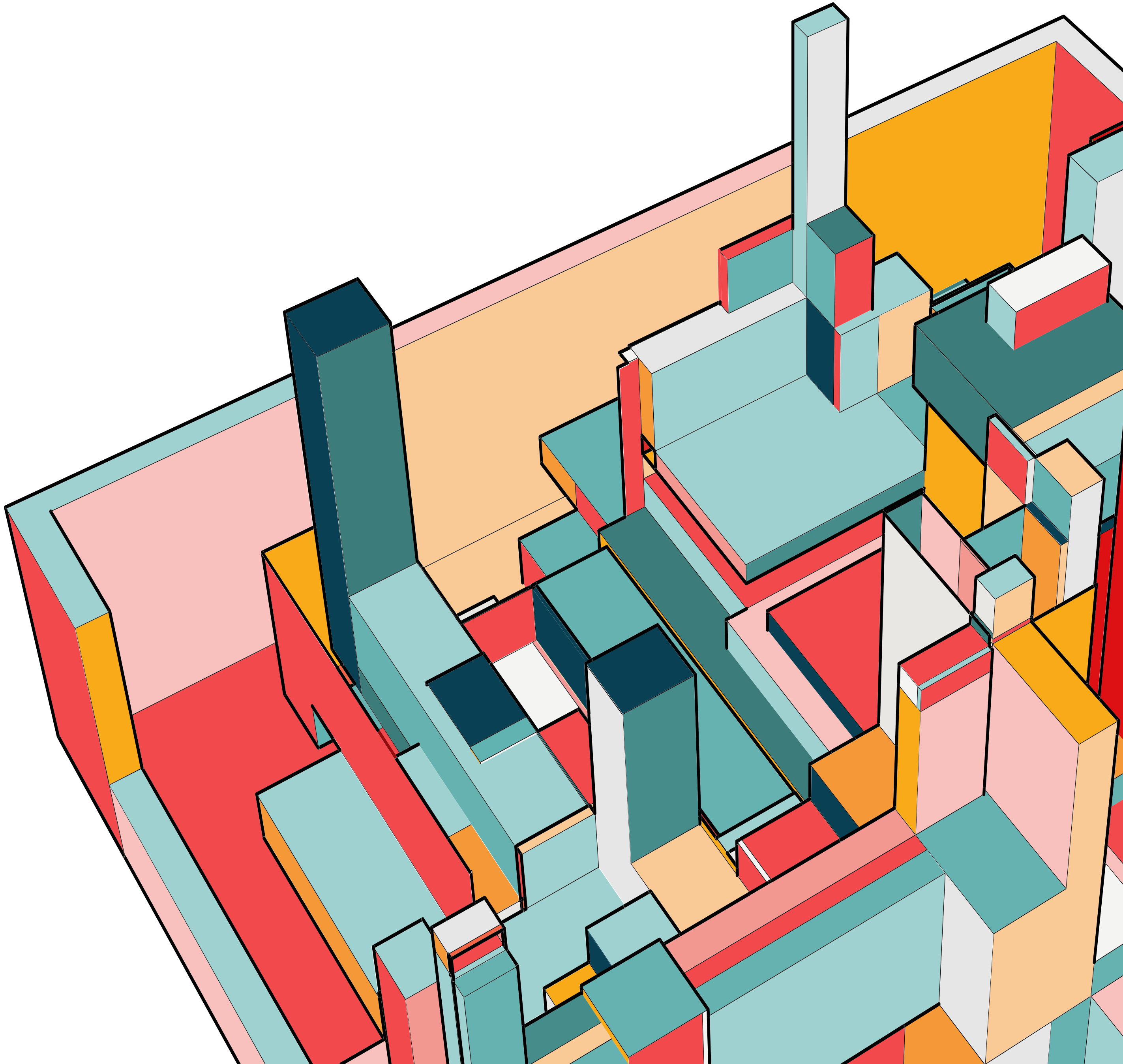
Neutral Colors

Neutral colors are a great selection to mix with cool or warm palettes. They are great for background in design, and tend to tone down the use of other bold colors. Black is added to create a 'shade', while white is added to create a 'tint'.

100% CONFIDENT

Your journey in this course is enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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